

Molecular Systems in Chemical Engineering –Spring 2022

Student Learning Outcome 1

Rubric for assessment problem on midterm exam (Problem #11)

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems
3. Appropriate engineering interpretation of mathematical and scientific terms	Shows appropriate engineering interpretation of mathematical and scientific terms	Most mathematical terms are interpreted correctly	Mathematical terms are interpreted incorrectly or not at all
4. Translates academic theory into engineering applications	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality	Some gaps in understanding the application of theory to the problem and expects theory to predict reality	Does not appear to grasp the connection between theory and the problem
6. The uses of appropriate resources needed to solve problems	Uses appropriate resources to locate information needed to solve problems	Uses limited resources to solve problems	Uses no resources to solve problems

Averages:

Performance Indicators:

1	2	3	4	6
1.75	1.75	1.92	2.08	2.00

Average Score	1.90
Standard dev.	0.94

PI 1/2 demonstrated the lowest level of performance. A significant number of students had difficulty interpreting the question and determining what was being asked. Consequently, many students chose not to attempt it. Upon further review, students indicated that the question was, in fact, approachable. However, it required multiple steps and calculations, while being assigned only 3 points out of 100. In future iterations of

the course, greater weight has been allocated to questions requiring higher levels of effort, while still maintaining appropriate value for more straightforward questions.

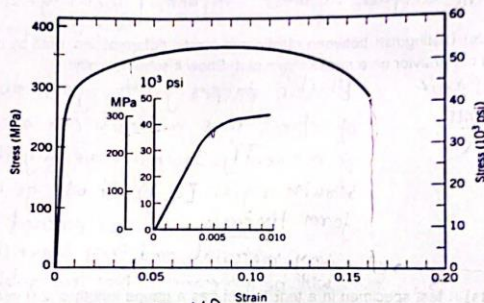
PI 4/6 demonstrated the highest level of performance. Students who attempted the question were generally successful in identifying the appropriate equations. However, several students stopped at listing the equations and did not proceed to substitute values or complete the calculations. To support student success, future offerings of the course permit the use of a cheat sheet and provide an equation sheet. While these resources are available, students are still responsible for selecting and correctly applying the appropriate formulas to solve the problem.

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11. (3 points) Given the engineering stress-engineering strain diagram for an aluminum alloy initially 11 mm in diameter and 50 mm in length, calculate the following:

- Modulus of elasticity
- Approximate the Yield strength
- Approximate the Tensile strength
- Percent elongation at the moment the aluminum part fractures
- True stress and true strain at an engineering strain of 0.04
- Elongation (ΔL) when a load of 25 kN is applied

Remember $1 \text{ Pa} = 1 \text{ N} \cdot \text{m}^{-2}$



A.] Young's Modulus = $m = \frac{250 \text{ MPa}}{0.005} = 50 \text{ GPa}$

B.] Yield stress from graph, can be given as 250 MPa.

C.] Tensile stress = The highest amount of stress while going through elongation = 360 MPa.

D.] % elongation = $\frac{0.163}{0.005} \times 100 = 16.3\%$

E.] $E_t = \ln\left[\frac{L}{L_0}\right] = \ln(1 + \epsilon) = \ln(1 + 0.04) = 0.03922$

$\sigma_t = \sigma(1 + \epsilon) = 340 \text{ MPa}(1 + 0.04) = 353.6 \text{ MPa}$

F.] $\sigma = \frac{F}{A_0} = \frac{25 \text{ kN}}{\pi\left(\frac{11}{2}\right)^2} = 263.07 \text{ MPa} \rightarrow \text{look at the graph.}$

G.] ΔL (elongation) by $F = 25 \text{ kN}$ would be $0.005 \times 50 \text{ mm}$
 $\Delta L = 0.25 \text{ mm}$